

Curriculum Vitae

WEI-CHE HSU

+1(623)373-1888 | weicheh1@asu.edu | 

EDUCATION

Arizona State University (ASU)

Doctor of Philosophy in Industrial Engineering

Aug 2026 – Present

Tempe, Arizona

- Fulton Scholar IV Graduate Research Associate

National Yang Ming Chiao Tung University (NYCU)

Master of Science in Industrial Engineering and Management

Sep 2021 – Aug 2023

Hsinchu, Taiwan

- Total Credits Earned: 37 | GPA: 3.96/4.3

National Tsing Hua University (NTHU)

Bachelor of Science in Industrial Engineering and Engineering Management

Sep 2017 – Jun 2021

Hsinchu, Taiwan

- Total Credits Earned: 164 | GPA: 3.46/4.3
- Acquired Minor in Power Mechanical Engineering & Data Science Interdisciplinary Program Certification

RELEVANT COURSEWORK

Optimization & Operations Research: Operations Research I & II; Advanced Operations Research (*simulation optimization and algorithm evaluation*); Linear Programming (*simplex methods, duality, and sensitivity analysis*); Integer Programming & Network Analysis (*integer formulations and network-flow models*); Stochastic Programming (*scenario-based optimization under uncertainty*)

Stochastic Modeling & Statistics: Probability Theory; Engineering Statistics; Stochastic Processes (*Markov chains and probabilistic system modeling*)

Data Analytics & Computing: Data Structures; Big Data Analytics & Data Mining (*data-mining methods and R-based analysis*); Smart Data Analytics (*analytical methods for engineering and healthcare applications*)

Mathematical Foundations: Linear Algebra; Discrete Mathematics; Numerical Analysis (*approximation methods, error analysis, and numerical stability*)

Additional Engineering Coursework: Engineering Mathematics I (*ordinary differential equations*); Manufacturing Processes; Applications of the Finite Element Method (*finite-element modeling and engineering analysis*)

WORK EXPERIENCE

Micron Technology – Industrial Engineer

Full-time, Department of Operations Intelligence

Mar 2024 – Jun 2026

Taichung, Taiwan

- Improve the transfer efficiency of the Automatic Material Handling System (AMHS)
 - Optimize carrier routing by balancing transfer efficiency and redundant movement under uncertainty
 - Automate the data collection process and enhance system visibility to support monitoring and analysis
- Conducted a project to optimize tool-step assignment strategies aimed at reducing inter-fab transfers under high loading conditions
 - Developed a simulation model to analyze how alternative assignment patterns influence transfer volume, given capacity limits, process-rate variability, and tool-specific step eligibility
 - Applied a heuristic search method to generate improved assignment configurations and identify path-release strategies that reduce inter-fab congestion without affecting the required wafer output

Micron Technology – Industrial Engineer

Intern, Department of Operations Intelligence

Jul 2022 – Aug 2022

Taichung, Taiwan

- Focused on analyzing AMHS cross-fab transfer causes
 - Identified key factors contributing to transfer inefficiencies and proposed actionable solutions
 - Designed interactive Tableau dashboards to automate data visualization

Vanguard International Semiconductor – Data Analytics Engineer

Intern, Department of Production Controlling

Jul 2021 – Aug 2021

Hsinchu, Taiwan

- Constructed production output forecasting models using big data techniques

TEACHING EXPERIENCE

Teaching Assistant — Course: Data Structure (Class Size: 60 Students) Feb 2023 – June 2023
Department of Industrial Engineering and Management, NYCU *Hsinchu, Taiwan*

- Provided student support on algorithm design, data structures, and programming concepts
- Designed homework and exam questions aligned with course learning objectives
- Graded coding assignments for 60 students and developed an automated grading system using predefined input/output cases to ensure consistent and accurate evaluation

RESEARCH EXPERIENCE

A Simulation-Based Opt. Study of Procurement Lot Sizing under Demand Uncertainty
M.S. Thesis (2023), Department of Industrial Engineering and Management, NYCU

- Studied a hypothetical raw material procurement problem with demand uncertainty to explore how stochastic factors affect ordering and inventory decisions
- Developed a simulation-optimization framework that combines heuristic and gradient-based optimization methods, achieving over a 10% improvement in solution quality on large-scale problems compared with using either approach alone

Energy Efficiency Opt. of Variable-Freq. Strategy for AC Systems in Wafer Mfg. Plant
Academia-Industry Collab. (2020), Department of Industrial Engineering and Engineering Management, NTHU

- Developed a data-driven frequency control strategy for air-conditioning systems in wafer manufacturing
- Built regression models and applied genetic algorithm (GA) & particle swarm optimization (PSO) to minimize energy consumption, which saved 70k US dollars per month for one of the SAS wafer's foundry

HONORS & AWARDS

Fulton Scholar IV, Ira A. Fulton Schools of Engineering, ASU	2026-27
Third Place, Summer Intern Project Competition, Vanguard International Semiconductor Corp. (VIS)	2021
Industrial Engineering Undergraduate Project Award, NTHU	2020

PROGRAMMING & ANALYTICAL TOOLS

Programming Languages: C/C++, C#, Python, R, MATLAB, JavaScript
Database: Microsoft SQL Server
Visualization: Tableau, Microsoft Power BI